

KUUIJUARAPIK, QC, Canada

WMO: 719050

Lat: 55.282N

Lon: 77.765W

Elev: 40

StdP: 14.67

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-32.3	-27.5	-40.4	0.5	-32.2	-35.7	0.7	-27.4	28.7	1.8	26.2	0.6	7.6	130	0.877

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.8	77.9	62.2	73.0	60.5	68.8	58.9	65.0	73.2	62.5	69.8	60.2	67.5	13.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.8	82.7	68.4	59.3	75.5	66.4	56.7	68.7	63.5	30.1	73.5	28.3	69.8	26.6	67.5	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.4	24.9	22.1	DB	-40.0	85.9	5.3	3.9	-43.8	88.7	-46.9	91.0	-49.8	93.2	-53.7	96.0	(4)
(5)			WB	-40.2	69.4	5.2	3.3	-43.9	71.8	-47.0	73.8	-49.9	75.7	-53.7	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.9	-6.1	-7.6	3.9	19.8	35.7	46.0	53.3	54.5	47.6	38.5	25.6	9.4		
	DBStd	24.05	12.81	12.55	15.05	12.30	10.79	10.12	8.32	7.34	6.70	6.62	7.65	12.72		
	HDD50	8918	1739	1614	1430	905	469	199	56	29	125	363	732	1257		
	HDD65	13938	2204	2034	1895	1355	909	577	377	339	523	821	1182	1722		
	CDD50	490	0	0	0	0	26	78	158	168	54	6	0	0		
	CDD65	36	0	0	0	0	1	6	14	14	1	0	0	0		
	CDH74	326	0	0	0	0	31	68	119	102	6	0	0	0		
	CDH80	68	0	0	0	0	9	13	28	18	0	0	0	0		
	WSAvg	10.6	10.6	9.7	9.6	10.0	9.6	9.5	9.1	10.0	12.2	12.0	12.8	12.6		
	PrecAvg	26.1	1.0	0.8	1.0	0.8	1.5	2.4	3.2	3.4	4.4	3.2	2.5	1.9		
(15) Precipitation	PrecMax	30.3	2.2	1.7	2.5	1.4	2.8	5.4	6.2	7.4	6.8	5.3	3.9	3.3		
	PrecMin	22.4	0.4	0.3	0.2	0.3	0.4	0.5	1.1	1.2	2.2	1.8	1.6	0.7		
	PrecStd	2.1	0.5	0.4	0.6	0.3	0.7	1.3	1.5	1.5	1.3	0.9	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.2	32.4	43.6	53.7	78.5	82.1	82.9	82.1	73.4	62.7	45.9	35.5		
		MCWB	31.5	30.5	36.7	43.0	58.7	63.3	63.7	65.4	62.7	55.3	43.4	33.9		
	2%	DB	24.2	24.2	38.0	46.4	67.4	74.3	77.0	76.4	67.5	56.9	40.1	32.3		
		MCWB	23.0	22.7	34.3	39.2	53.2	59.7	62.0	63.0	59.3	52.6	37.8	31.3		
	5%	DB	19.2	18.8	32.7	42.0	59.8	68.8	72.1	71.3	62.7	52.0	37.2	29.3		
		MCWB	18.3	17.5	30.3	36.5	49.2	56.8	60.3	61.3	56.3	47.7	35.7	28.2		
	10%	DB	12.9	12.0	27.1	37.7	53.1	63.7	67.2	66.8	58.4	48.0	35.0	26.6		
		MCWB	11.9	10.8	24.9	33.8	44.8	53.8	58.8	59.5	53.6	44.7	33.4	25.4		
	0.4%	WB	31.9	30.8	37.8	44.1	59.5	67.5	67.9	67.8	64.7	56.7	43.6	34.2		
		MCDB	32.1	32.5	42.1	51.5	77.1	76.2	77.1	76.9	70.7	60.7	45.1	35.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	23.2	22.7	34.5	40.2	54.0	62.3	64.0	65.1	60.5	52.7	38.3	31.3		
		MCDB	23.9	24.1	37.8	45.5	65.0	71.9	71.6	73.8	66.2	56.1	39.6	32.1		
	5%	WB	18.2	17.5	30.2	37.0	49.7	57.9	61.8	62.6	56.9	48.3	35.8	28.4		
		MCDB	19.1	18.7	32.7	41.2	59.2	67.0	69.6	69.1	61.4	51.6	37.1	29.3		
	10%	WB	12.0	10.8	24.8	33.8	44.9	54.1	59.5	59.9	53.6	44.8	33.7	25.4		
		MCDB	12.8	12.0	27.1	37.4	52.7	63.2	67.2	66.3	58.4	47.7	34.9	26.6		
	5% DB	MDBR	13.7	17.1	18.4	15.9	14.8	16.9	15.6	12.8	11.0	8.7	8.4	11.1		
		MCDBR	15.3	20.2	20.6	20.9	27.6	26.9	24.3	21.5	19.6	15.2	8.5	10.0		
		MCWBR	15.3	19.6	18.1	15.0	16.1	15.3	13.5	11.4	13.3	11.7	7.9	10.0		
		MCWBR	15.5	20.2	20.0	18.8	26.9	25.7	21.4	18.9	18.1	14.4	8.5	9.9		
(35) Mean Daily Temperature Range	5% WB	MCWBR	15.5	19.6	17.8	14.0	16.5	17.3	14.1	11.6	13.9	12.2	8.1	10.0		
	taub		0.222	0.247	0.284	0.295	0.318	0.342	0.356	0.364	0.351	0.320	0.281	0.214		
		taud	2.203	2.201	2.141	2.164	2.285	2.420	2.487	2.461	2.468	2.472	2.260	2.221		
(40) Clear-Sky Solar Irradiance	Ebn at Noon		234	268	281	294	293	286	280	270	258	239	201	211		
		Edh at Noon	17	26	36	40	38	34	31	30	26	21	17	14		
	RadAvg		251	572	1042	1521	1729	1796	1613	1243	787	431	177	141		
(44) All-Sky Solar Radiation	RadStd		27	52	110	140	148	184	144	81	78	34	23	20		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+2.27	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)		N/A	N/A	+2.09	-1.30	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page